18.1 Fraction Multiplication - Worksheet 1

1 Draw a diagram to represent the product $\frac{1}{4} \cdot \frac{3}{2}$.

2 Draw a diagram to represent the product $\frac{2}{3} \cdot \frac{3}{4}$.

3 Calculate $\frac{4}{5} \cdot \frac{8}{3}$.

4 Calculate $\frac{4}{5} \cdot \frac{8}{3}$.

18.2 Fraction Multiplication - Worksheet 2

1 Draw a diagram to represent the product $\frac{3}{4} \cdot \frac{3}{4}$.

2 Calculate $\frac{10}{9} \cdot \frac{4}{7}$.

3 Calculate $\frac{12}{5} \cdot \frac{15}{8}$.

4 Calculate $\frac{3a}{8} \cdot \frac{5}{7b}$.

18.3 Fraction Multiplication - Worksheet 3

1 Calculate $\frac{3}{11} \cdot \frac{8}{5}$.

2 Calculate $\frac{25}{6} \cdot \frac{8}{3}$.

3 Calculate $\frac{5x}{8y} \cdot \frac{7x^2}{9}$.

4 Calculate $\frac{15p^2}{7q} \cdot \frac{21pq}{5}$.

18.4 Fraction Multiplication - Worksheet 4

1 Check the presentation for errors. If you find one, circle it and describe the mistake in words.

$$\frac{3}{4} \cdot \frac{4}{3} = \frac{\cancel{3}}{\cancel{4}} \cdot \frac{\cancel{4}}{\cancel{3}} \qquad \text{Reduce}$$
$$= 0$$

2 Calculate $\frac{3}{5} \cdot 5x$.

Can you explain why $5x = \frac{5x}{1}$? Can you think of another way of understanding this calculation?

3 Calculate $\frac{15x^2y}{8z} \cdot \frac{10xz^2}{9y}$.

4 Calculate $\frac{5(x+2)^2}{8(x-3)} \cdot \frac{4(x-3)^3}{15(x+2)^4}$.

Keep the terms in the parentheses together. Do not use the distributive property to expand those terms.

18.5 Fraction Multiplication - Worksheet 5

1 Check the presentation for errors. If you find one, circle it and describe the mistake in words.

$$\begin{aligned} \frac{5}{14} \cdot \frac{7}{15} &= \frac{5}{2 \cdot 7} \cdot \frac{7}{3 \cdot 5} && \text{Identify common factors} \\ &= \frac{\cancel{5}}{2 \cdot \cancel{7}} \cdot \frac{\cancel{7}}{3 \cdot \cancel{5}} && \text{Reduce} \\ &= 6 \end{aligned}$$

2 Calculate $\frac{15p^2}{7q} \cdot \frac{21pq}{5}$.

3 Calculate $\frac{x^2-1}{x^2-x-6} \cdot \frac{x^2-5x+6}{x+1}$.

Factor the numerator and the denominator.